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#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <math.h>

int main(){
    // Each time take an input n and check if it is a multiple of 11
    int i = 0, j = 0, l = 0, sum_odd = 0, sum_even = 0, diff = 0, flag = 0;
    char arr[1001], c = 0;

    while(1)
    {
        // each time take n as input into array
        i = 0;
        while (1)
        {
            c = 0;
            scanf("%c", &c);
            if(c == '\n')
                break;
            arr[i] = c;
            i++;
        }
        arr[i] = '\0';
        // check for end of input
        if (arr[0] == '0')
        {
            break;
        }

        l = strlen(arr), sum_odd = 0, sum_even = 0;
        //compute sums
        for (j = 0; j < l; j++)
        {
            if (j % 2 == 0) {

                sum_even += arr[j] - '0';
            }
            else
            {
                sum_odd += arr[j] - '0';
            }
        }
        //compute difference sums
        diff = abs(sum_odd - sum_even);

        // check if it's a multiple of 11
        if (diff == 0 || diff % 11 == 0)
        {
            printf("%s is a multiple of 11.\n", arr);
        }
        else {
            printf("%s is not a multiple of 11.\n", arr);
        }
    }
}

```